11nix Command Reference

11nix 0.1 on the PDP-11/23 with a QBone · DEC MMU, DEQNA/DELQA and QBone port (c) 2026 N9TAX RetroLAB · based on Fuzix

Starting and stopping

START, then 10000g	Boot: press START in the QBone web panel. At the ODT prompt @ on the console type **10000g**. 11nix boots to a # prompt: you are root.
halt	**Stop the machine properly.** Writes everything to the disk, marks it clean, and halts the processor. **reboot** does the same.
START, 10000g again	Boot again after a halt. Always START first: the kernel overwrites the boot loader, so **10000g** alone does not work.
sync	Write buffered data to the disk now. Not a shutdown: the disk stays marked in use.

Read-only disk? If the machine stopped any other way (HALT switch, power, crash), the next boot prints **WARNING: the disk is READ-ONLY**. Nothing can be saved, and the network does not start. Repair it: **fsck /dev/hd0**, answer **y** to every question (inodes "not of correct type" are old network sockets: zap them). fsck halts the machine when it is done: START, **10000g**.

Network setup (once)

There is no DHCP: the address is in **/etc/netrc**, the DNS server in **/etc/resolv.conf**. Example for a LAN 10.1.1.x with the router at 10.1.1.1:

```
# cat > /etc/netrc
ipaddr  10.1.1.80
gateway 10.1.1.1
netmask 255.255.255.0
                              <- press Ctrl-D here
# echo nameserver 10.1.1.15 > /etc/resolv.conf
# halt                        <- then START, 10000g
```

At boot it prints **network: inet 10.1.1.80 ...**. From another computer: **telnet 10.1.1.80** (log in as **root**), or **http://10.1.1.80/** in a browser. The BeagleBone itself cannot reach the PDP-11 over the DELQA; every other machine can.

Keys at the prompt

Ctrl-C	Interrupt the running program.
Ctrl-D	End of input (finishes **cat > file**).
Backspace	Erase a character (Ctrl-H and DEL both work). Ctrl-U erases the whole line.
Ctrl-S / Ctrl-Q	Pause / resume output. Ctrl-O throws output away until pressed again.
Arrow keys	Not at the prompt: the shell has no line editing or history. An arrow only moves the cursor on the screen and puts junk in the line: Ctrl-U, and type it again.

The shell (V7 Bourne shell)

cmd > file	Output to a file (**>>** appends). **< file** reads from a file.
cmd1 | cmd2	Pipe. Careful: **^** is also a pipe in this shell, so quote it: **'^?'**.
cmd &	Run in the background; **ps** shows it, **kill PID** stops it.
cmd1 ; cmd2	One after the other.
*** ? [abc]**	Wildcards: **ls /bin/c***.
`cmd`	Insert the output of a command: **echo today is `date`**.
'text' "text"	Quotes. Inside **"..."**, **$VAR** is still replaced.
VAR=value	Set a variable; **export VAR** passes it to programs. **$PATH** is **/bin**.
sh file	Run a shell script. **if/then/else/fi**, **for x in ...; do ...; done**, **while** work as in any Bourne shell.

cd dir, pwd	Change / show the directory. There is no **~**, history or tab completion.
exit	Leave the shell (logs out a telnet session).

Files and directories

ls -l [dir]	List files, long form (**ls -a** shows hidden ones).
cat file	Show a file. **more file** shows it a page at a time.
head / tail file	First / last 10 lines (**head -20 file**, **tail -n 20 file**).
cp a b, mv a b	Copy / move or rename.
rm file	Delete. **rmdir dir** deletes an empty directory.
mkdir dir	Make a directory.
ln a b	A second name for a file (**vi** is a link to **levee**).
touch file	Create an empty file, or update its time.
chmod 755 file	Permissions. **chown** / **chgrp** change the owner / group.
du [dir], df	Disk used by files / free space on the disk.
od -c file	Dump a file byte by byte. **dd** copies raw blocks.
tar -cvf x.tar dir	Archive a directory; **tar -tvf x.tar** lists, **tar -xvf x.tar** unpacks.
cmp a b	Compare two files. **sum** / **cksum** give checksums.
rx file, sx file	XMODEM receive / send over the console (from your terminal program).

Text

grep word file	Lines containing a pattern (**-v** the others, **-i** ignore case). **fgrep** for plain text.
sed s/old/new/ file	Replace text in a stream.
sort, uniq	Sort lines; drop repeated lines.
wc file	Count lines, words, characters.
tr a-z A-Z	Translate characters.
cut -c1-10 file	Columns of each line (**-d: -f1** for fields).
tee file	Copy a pipe into a file too.
xargs cmd	Run a command with the words read from input.
echo text	Print text. **banner HI** prints it big.

System

ps	Running processes. **kill PID**, **kill -9 PID**, **killall name**.
free	Memory in KB: 176 KB for programs, 14 processes at most, the rest swapped to disk.
uptime, uname -a	Time since boot; system name and version.
date	The time (UTC). It is set from the internet at boot; **setdate** sets it by hand.
mount	Mounted disks. **/dev/hd0** is the RL02.
fsck /dev/hd0	Check and repair the disk (see Starting and stopping).
stty	Terminal settings (**stty -a** shows them).
passwd	Set root's password (none at first: anyone on the LAN can telnet in).
su, id, who, whoami	Change user; who you are; who is logged in.
which cmd	Where a program is.

Network

ifconfig	Address, netmask, gateway and the Ethernet address of the DELQA.
ping host	Ping a name or address until Ctrl-C.
dig name [@server]	Look up a name in DNS.
telnet host [port]	Connect to another machine (**-a** passes ANSI codes through).
htget http://host/path file	Fetch a web page into a file (http only, no https).
ntpdate -s server	Set the clock (UTC). **-o -6** adds an offset in hours, for example **ntpdate -s -o -6 pool.ntp.org** for US Central standard time.
tinyirc server:port [nick]	A small IRC client.
echoping addr [port] [msg]	Send a UDP echo.

Started at boot by **/etc/rc**: **netd** (TCP/IP, reads **/etc/netrc**), **telnetd** (port 23), **httpd** (port 80, serves **/www**), **ntpdate**. Edit **/etc/rc** to change what starts, for example the time zone offset of ntpdate.

ed, the line editor (works on any terminal)

Type a command and Enter. Lines are numbered from 1; **$** is the last line.

ed file	Start editing a file.
1,$p	Print all lines. **3p** prints line 3.
/text/	Go to the next line containing text.
3c or /text/c	Replace that line: type the new text, then **.** alone on a line.
$a	Add lines after the last one; finish with **.** alone. **3i** inserts before line 3.
3d	Delete line 3. **2,4d** deletes lines 2 to 4.
s/old/new/	Replace text in the current line (**1,$s/old/new/g**: everywhere).
w	Save.
q	Quit (asks "Really quit?" when the changes are not saved).

vi (levee), the screen editor (VT100 / ANSI terminal)

vi file (or **levee file**). It starts in **command mode**: keys are commands. **i**, **a** or **o** switch to typing; **Esc** switches back. It edits files up to about 4 KB.

arrows or h j k l	Move left, down, up, right. **0** / **$** start / end of line, **G** last line, **1G** first. While typing, an arrow ends the typing (like Esc) and then moves.
i a A o	Type before the cursor / after it / at the end of the line / on a new line below. Backspace erases what you typed. Then Esc.
x	Delete the character under the cursor.
dd	Delete the line. **3dd** deletes 3 lines.
cw	Change a word: type the new one, then Esc.
r c	Replace one character with c.
yy p	Copy the line; put it below.
u	Undo.
/text	Search; **n** finds the next one.
:wq or ZZ	Save and quit.
:w :q!	Save / quit without saving.

Files to know

`/etc/netrc`	This machine's IP address, gateway, netmask.
`/etc/resolv.conf`	The DNS server: **`nameserver 10.1.1.15`**.
`/etc/rc`	Run at boot: starts the network, telnetd, httpd, ntpdate.
`/.profile`	Run when the console shell starts (your **`stty`** settings).
`/www/index.html`	The home page of the web server.
`/usr/lib/basic/*.bas`	Examples for **`ubasic /usr/lib/basic/demo.bas`**.

Fun

`/usr/games/advent`	Colossal Cave Adventure. **`quit`** to stop.
`ubasic file.bas`	Integer BASIC.
`cal 11 2026`	A calendar. **`factor 1170`** prime factors, **`seq 1 10`** numbers.